

KARTIK TRIPATHI

DATA ANALYST | SQL & BUSINESS INTELLIGENCE DEVELOPER

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PROFESSIONAL SUMMARY

Analytical Data Analyst certified with the **IBM Data Analytics Professional Certificate** and holding a B.Sc. in Physics, Chemistry, & Mathematics. Rigorous expertise in **SQL (Advanced Queries, DDL/DML, Window Functions)**, **Python (Pandas, NumPy, SciPy)**, **Power BI, Tableau, Advanced Excel, and Applied Statistics**. Proven ability in architecting normalized relational databases, developing executive BI dashboards, and executing predictive customer RFM segmentations to drive business decision-making.

TECHNICAL SKILLS

- **SQL & Databases:** MySQL, SQLite, PostgreSQL, Relational Schema Normalization (3NF), DDL/DML, Window Functions (ROW_NUMBER), Joins, Subqueries (EXISTS), Set Operations
- **Programming & Analytics:** Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn, SQLAlchemy), Jupyter, RFM Segmentation, Pareto (80/20) Decile Modeling, Data Quality Profiling
- **BI & Visualization:** Microsoft Power BI (DAX Measures, Power Query ETL, Star Schema, KPI Dashboards), Tableau Desktop (Calculated Fields, LOD Expressions, Geo-Maps)
- **Spreadsheets & Stats:** Advanced Excel (Multi-Table Pivot Tables, Slicers, XLOOKUP, Nested IFS/COUNTIFS, Variance Modeling), Hypothesis Testing (t-Tests, ANOVA, Chi-Square, Pearson r)

FEATURED TECHNICAL PROJECTS

Retail E-Commerce Relational Database & Advanced SQL Analytics | *MySQL, Relational Schema Design (3NF), Window Functions, Joins*

- **Schema Architecture & Integrity:** Designed and implemented a normalized 6-table relational database (customers, products, orders, order_items, payments, reviews) with primary/foreign keys, CHECK constraints, and automated timestamps.
- **Advanced Query Analytics:** Authored a 6-tier analytical query suite utilizing multi-table INNER/LEFT/RIGHT JOINS, correlated subqueries (EXISTS/NOT EXISTS), and window functions (ROW_NUMBER() OVER PARTITION) for customer order ranking and gateway settlement reconciliation.

AegisLife Insurance: Risk Analytics & Claims Intelligence (Capstone) | *Python (Pandas, SciPy), SQLite, SQL DDL, Power BI*

- **Relational ETL & Statistical Rigor:** Ingested 2M+ records into normalized SQLite schema; ran parametric hypothesis tests in SciPy disproving legacy Risk Score predictive power ($r = 0.012$, $p = 0.64$) to recommend an underwriting model overhaul.
- **Fraud Detection & Power BI:** Isolated intermediary agent fraud clusters ($\chi^2 = 38.42$, $p < 0.001$) and delivered an interactive Power BI dashboard with C-suite briefing decks to mitigate a 49% claims suspicion rate.

Customer Lifetime Value & RFM Segmentation Analytics Pipeline | *Python (Pandas, Seaborn), Jupyter, Pareto Decile Modeling*

- **RFM Quantile & Cohort Modeling:** Processed 23,050 orders across 1,000 accounts into 8 behavioral cohorts via pandas qcut quintile scoring; classified Champions (\$3.73M/16.2% rev) and Loyalists (\$5.12M/22.2% rev) for targeted lifecycle marketing.
- **Dual-Axis Pareto Revenue Analysis:** Built a dual-axis Pareto distribution chart mapping decile spend against cumulative revenue share, confirming top 20% of spenders generate \$6.23M (27.0%) and top 70% generate 78.3% of total revenue.

Enterprise Business Intelligence & Advanced Analytics Dashboards | *Tableau Desktop, Power BI (DAX), Advanced Excel*

- **Tableau Real Estate Analytics:** Analyzed 12 years of property transactions, mapping \$9.6B+ sales in Hot Springs and calculating Assessed-to-Sale ratios (1.08-1.11 benchmarks) for institutional discount valuation.
- **Power BI & Excel Modeling:** Engineered an Airline delay profitability dashboard in Power BI (quantifying delays >25 mins cause 18.4% profit erosion) and an Amazon multi-table sales quota variance model in Excel.

CERTIFICATIONS & EDUCATION

IBM Data Analytics Professional Certificate: SQL & Databases, Python, Statistics, Excel, Power BI, Tableau, Data Transformation, Generative AI.
Bachelor of Science (B.Sc.) in Physics, Chemistry, & Mathematics | Kumaun University, Uttarakhand, India (2020)